

Multi Threading And Multi Processing

- ① Program
- ② Process
- ③ Threads
- ④ Examples of Process And Thread.

① Program → A program is a sequence of Instructions written in programming language [eg: Python, C++]

Eg: Google Chrome → .exe ← Program → Browser should work.

Word

Exec

② Process : A process is simply an instance of a program that is being executed

works along with OS so need some resources in the memory

each process runs will have, a

Separate Memory Space

① One process cannot corrupt another process



for dynamic allocation

Eg: Browser

Exec → process → Id

App → Process.

③ I/O Requirements [Resource]

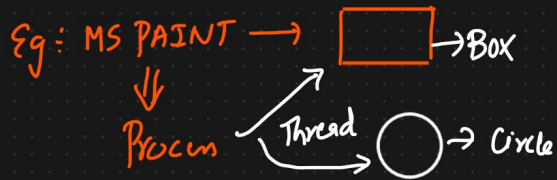
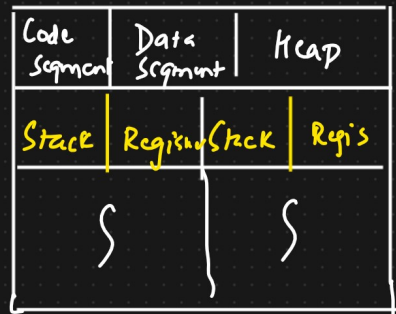
① → Increased execution time to switch between processes.

③ Thread → A thread is a unit of execution within a process

Single Threaded Process



Multi Threaded Process



all threads share the same memory space and code, data, heap